

● EASY

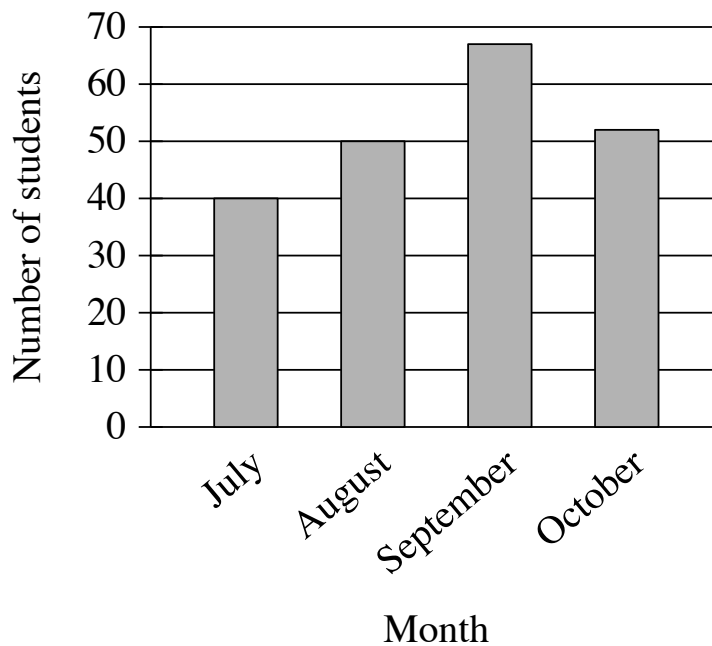
● PROBLEM-SOLVING AND DATA ANALYSIS

Problem-Solving and Data Analysis

15

10 questions · ~15 min

The bar graph shows the distribution of the number of students from one school who were born in one of four months.



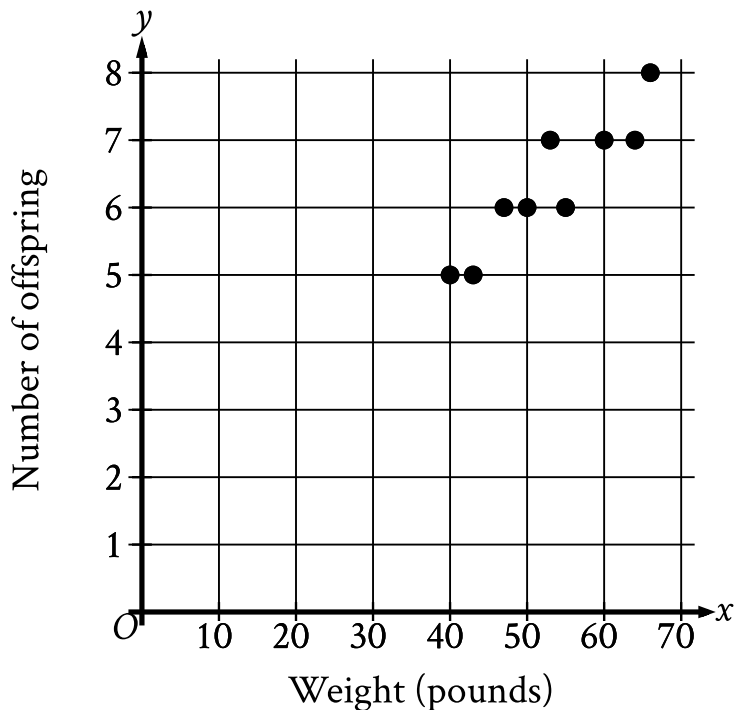
- The Number of students data for the 4 bars are as follows:
 - July: 40
 - August: 50
 - September: 67
 - October: 52

How many more students were born in August than were born in July?

- A. 90
- B. 50
- C. 40

D. 10

The scatterplot shows the relationship between the weight, in pounds, of each of 9 female gray wolves on April 30 and the number of offspring each gray wolf produced.



- The scatterplot has 9 data points.
- The data points are in a linear pattern trending up from left to right.
- The data points have the following coordinates:
 - (40 comma 5)
 - (43 comma 5)
 - (47 comma 6)
 - (50 comma 6)
 - (53 comma 7)
 - (55 comma 6)
 - (60 comma 7)
 - (64 comma 7)
 - (66 comma 8)

How many offspring did the 50-pound gray wolf produce?

A. 8

B. 7

C. 6

D. 5

Q3

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

How many yards are equivalent to 77 rods? 5.5 yards = 1 rod

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

PROBLEM-SOLVING AND DATA ANALYSIS

Lani spent 15% of her 8-hour workday in meetings. How many minutes of her workday did she spend in meetings?

- A. 1.2
- B. 15
- C. 48
- D. 72

Q5

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

A list of 10 data values is shown.

6, 8, 16, 4, 17, 26, 8, 5, 5, 5

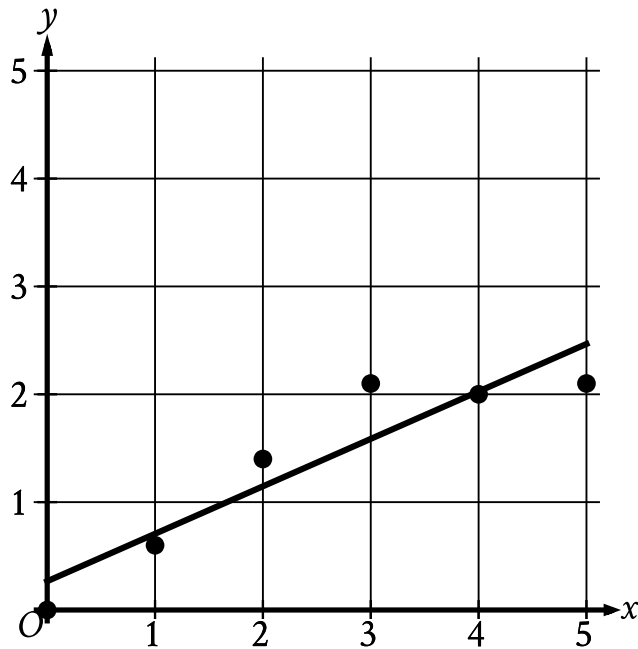
What is the mean of these data?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The scatterplot shows the relationship between x and y . A line of best fit is also shown.



- The scatterplot has 6 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown:
 - The line of best fit slants up from left to right.
 - 1 point is touching the line of best fit.
 - 2 points are above the line of best fit.
 - 3 points are below the lines of best fit.
 - The line of best fit passes through the following approximate coordinates:
 - (0 comma 0.2)
 - (4 comma 2)

Which of the following is closest to the slope of the line of best fit shown?

- A. -2.27
- B. -0.44
- C. 0.44
- D. 2.27

Q7

PROBLEM-SOLVING AND DATA ANALYSIS

To make a bakery's signature chocolate muffins, a baker needs 2.5 ounces of chocolate for each muffin. How many pounds of chocolate are needed to make 48 signature chocolate muffins? (1 pound = 16 ounces)

- A. 7.5
- B. 10
- C. 50.5
- D. 120

Each value in the data set shown represents the height, in centimeters, of a plant.

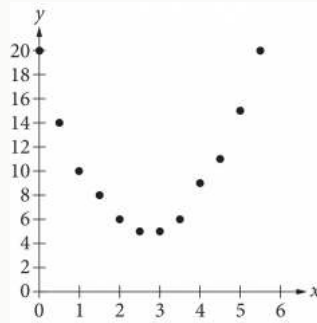
6, 10, 13, 2, 15, 22, 10, 4, 4, 4

What is the mean height, in centimeters, of these plants?

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



Of the following, which is the best model for the data in the scatterplot?

A. $y = 2x^2 - 11x - 20$

B. $y = 2x^2 - 11x + 20$

C. $y = 2x^2 - 5x - 3$

D. $y = 2x^2 - 5x + 3$

Q10

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

A mechanical device in a workshop produces items at a constant rate of 60 items per hour. At this rate, how many items will the mechanical device produce in 3 hours?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.